Finite Automata in Haskell

V. Velkey, W. Vromen

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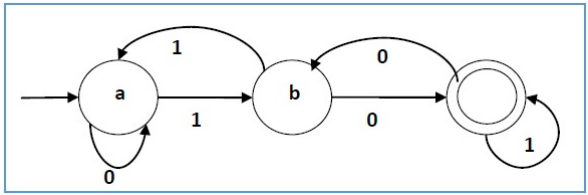
Overview

- Theory overview: DFA-s, NFA-s, ϵ-NFA-s, Regular expressions
- DFA -s: implementation, evaluation of words
- NFA-s and ϵ-NFA-s: implementation, evaluation
- Powerset construction: ϵ-NFA to DFA
- RegExp: implementation, word generation
- RegExp to DFA
- DFA to RegExp

Theory Introduction - Regular Languages

- Regular languages: "read-only computation"
  - Starting point of formal language theory
- Equivalent characterisations: Finite Automata, Regular expressions
- Goal of project: formalise their equivalence in Haskell

Theory - DFA-s



Formal definition: $(Q_n, \Sigma, \delta_n, q_0, F_n)$

- Q_n set of states
- Σ alphabet
- δ_n transition function (between states): $\delta(\mathrm{st}, \mathrm{sym}) = \mathrm{st}'$
- q_0 start state
- F_n set of accept states

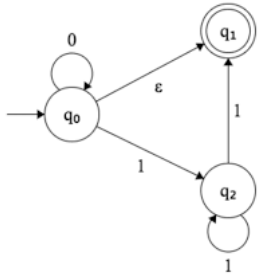
Theory - NFA-s, ϵ-NFA-s

NFA-s:

- can transition to multiple states
- Accept: if at least one path at accept

ϵ-NFA-s:

- Can transition between states without moving along the word



Theory - Regular expressions

Algebraic (recursive) way to build languages

- $\emptyset$, defines $L(\emptyset) = \emptyset$.
- ϵ, defines $L(\epsilon) = \{\epsilon\}$
- a, defines $L(a) = \{a\}$
- $E \cup F$, defines $L(E \cup F) = L(E) \cup L(F)$
- EF, defines $L(E)L(F)$ (concatenation)
- E^* defines $L(E^*) = L(E)^* = \bigcup_{n \geq 0} L(E)^n$

Theory - Equivalence

- For every ϵ-NFA, there is an equivalent DFA (powerset construction below)
- For every DFA there is an regular expression with the same language ($R_{i,j,k}$ construction below)
- For every RegExp, there is an ϵ-NFA on the same language (regExpToENFA)

DFA -s: implementation, evaluation of words

Design choice: words with unknown symbols will be *rejected*.

```
type Symbol = Char
data DFA a = DFA { states :: [a]
        , alphabet:: [Symbol]
        , delta :: Symbol -> a -> a
        ,  start:: a
        , acceptstate:: [a]}

evaluate:: (Eq a, Ord a) => DFA a -> String -> Bool
evaluate df st = all (`elem` alphabet df) st && -- all characters in
    string are in the alpahbet
                stateArr (start df) st `elem` acceptstate df where
                  stateArr  q [] = q
                  stateArr q (x:xs) = stateArr (delta df x q) xs
```

DFA Arbitrary

To generate functions: use idea from HW2

```
randomDelta :: (Eq a, Ord a, Arbitrary a) => [Symbol] -> [a] -> [a] -> Gen
     (Symbol -> a -> a)
randomDelta [] _ _ = return (const undefined)
randomDelta (sym:syms) ds ps = do
f <- randomDelta syms ds ps
wResult <- randomFunFromTo ds ps
return $ \sy -> if sy == sym then wResult else f sy

instance Arbitrary (DFA Int) where
arbitrary = do
-- choose a set of up to 4 worlds:
sts <- (0 :) <$> sublistOf [1..3]
let sym = ['0', '1']
delt <- randomDelta sym sts sts
strt <- elements sts
accraw <-  sublistOf sts
let acc = nub (0:accraw)
return $ DFA sts sym delt strt acc
```

NFA implementation

Transitioning between lists of states: double recursion.

```
data NFA a= NFA { statesNF :: [a]
        , alphabetNF:: [Symbol]
        , deltaNF :: Symbol -> a -> [a]
        , startNF:: a
        , acceptstateNF:: [a]}

evaluateNF:: Eq a => NFA a-> String -> Bool
evaluateNF nf st = all (`elem` alphabetNF nf) st && -- captures elements
    of string in alphabet
                any (`elem` acceptstateNF nf) (stateArrNF' [startNF nf] st
                    )  where
                  stateArrNF'  qs [] = qs
                  stateArrNF'  [] _ = []
                  stateArrNF'  [q] (x:xs) = stateArrNF' (deltaNF nf x q)
                      xs --recursion on string
                  stateArrNF'  (q:qs) (x:xs) = stateArrNF' [q] (x : xs) `
                      union` stateArrNF' qs (x : xs) --recursion on
                      statespace
```

ϵ-NFA-s

Design choice: ϵ-transitions represented as list of tuple - ease with transitive closure

```
data ENFA a = ENFA { statesENF :: [a]
        , alphabetENF:: [Symbol]
        , deltaENF :: Symbol -> a -> [a]
        , epTrans:: [(a, a)]
        , startENF:: a
        , acceptstateENF:: [a]}

rtClose:: (Eq a, Ord a) => ENFA a -> [a] -> [a]
rtClose nf qs = sort (unionL [[p | p <- statesENF nf , (q, p) `elem`
    trClose rcl ] | q<- qs] ) where
        rcl = refClose (statesENF nf) (epTrans nf)
```

The powerset construction - definition

- Take NFA $N = (Q_n, \Sigma, \delta_n, q_0, F_n)$
- Powerset construct: $D = (\mathcal{P}(Q_n), \Sigma, \delta, S, F_D)$
  - States: subsets of Q_n
  - Alphabet: same (of course)
  - transition function: $\delta_D(R, \sigma) = \cup_{s \in R} \delta_N(s, \sigma)$
  - Start state: singleton with previous start state
  - Accept states: $\{R \subset Q_N : S \cap F_n \neq \emptyset\}$
- For ϵ-NFA:
  Similar idea, but at each step, need to take ϵ-closure at steps
  - Start state $\mathrm{ecl}(\{q_0\})$
  - transition function: $\delta_D(R, \sigma) = \mathrm{ecl}(\cup_{s \in R} \delta_N(s, \sigma))$

Powerset construction - implementation

For ϵ-NFAs:

```
enfaToDFA:: (Eq a, Ord a) => ENFA a -> DFA [a]
enfaToDFA (ENFA sts alph del eps strt ac) = let nf = ENFA sts alph del eps
     strt ac in
cutDFA $ DFA sortedStates alph del' (rtClose nf (sort [strt])) [st | st <-
     sortedStates,  intersect st ac /= []] where
del' sy ls = let nf = ENFA sts alph del eps strt ac in rtClose nf ( unionL
     [del sy l | l <- ls] )
sortedStates = map sort $ subsequences sts
```

- Problem: By definition, exponential-size DFA
- Working with the output DFA (2^{19} states): unfeasible
- Workaround: cutDFA - only work with states reachable from start state
- Even better: construct *minimal* equivalent DFA
  - Not too efficient

RegExp implementation: data type

```
data RegExp = Empty | Epsilon | R [Symbol] | Union RegExp RegExp
            | Star RegExp | Con RegExp RegExp | Plus RegExp
  deriving (Show, Eq)
```

Design choices:

RegExp implementation: data type

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data RegExp = Empty | Epsilon | R [Symbol] | Union RegExp RegExp
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  deriving (Show, Eq)
```

Design choices:

- not split empty from non-empty expressions
- `Epsilon` versus `R []`
- `R [Symbol]` versus `R Symbol`

For convenience:

```
regExpUnion :: [RegExp] -> RegExp
regExpUnion = simplify . foldr Union Empty
```

RegExp but prettier

```
pRegExp :: RegExp -> String
pRegExp Empty = ""
pRegExp Epsilon = "R e"
pRegExp (R [])  = "R e"
pRegExp (R xs) = show xs
pRegExp (Union r1 r2) = "(" ++ pRegExp r1 ++ "|" ++ pRegExp r2 ++ ")"
pRegExp (Star r) = "(" ++  pRegExp r ++ ")*"
pRegExp (Con r1 r2) = pRegExp r1 ++ pRegExp r2
pRegExp (Plus r) = "(" ++ pRegExp r ++ ")+"

ppRegExp :: RegExp -> IO ()
ppRegExp = print . pRegExp
```

`ghci> ppRegExp (Con (Union (R "53") (R [])) (Star (R "19")))`

then becomes

`"("53"|R e)("19")*"`

RegExp Arbitrary

```
instance Arbitrary RegExp where
arbitrary = sized randomReg where
randomReg :: Int -> Gen RegExp
randomReg 0 = R <$> elements ["0", "1"]
randomReg n = oneof [ R <$> elements ["0", "1"]
, Star <$> randomReg (n `div` 8)
, Plus <$> randomReg (n `div` 8)
, Con <$> randomReg (n `div` 8)
<*> randomReg (n `div` 8)
, Union <$> randomReg (n `div` 8)
<*> randomReg (n `div` 8)
, return Epsilon ]
```

We chose `` `div` 8 `` to avoid dealing with monstrosities

RegExp to Generate Words

```
generateString :: RegExp -> Gen String
generateString = (`generateString'` 5) . simplify
        where generateString' :: RegExp -> Int -> Gen String
        generateString' Empty _ = error "cannot generate from empty
            language"
        generateString' Epsilon _ = return ""
        generateString' (R xs) _ = return xs
        generateString' (Union r1 r2) n = oneof [generateString' r1 n,
            generateString' r2 n]
        generateString' (Con r1 r2) n = do
                w <- generateString' r1 n
                v <- generateString' r2 n
                return (w ++ v)
        generateString' (Star _) 0 = return ""
        generateString' (Star r) n = oneof [generateString' Epsilon n,
            generateString' (Con r (Star r)) (n - 1)]
        generateString' (Plus r) n = generateString' (Con r (Star r)) n
```

Size and Error

```
generateString = (`generateString'` 5) . simplify
...
generateString' (Star r) n = oneof [generateString' Epsilon n,
            generateString' (Con r (Star r)) (n - 1)]
```

- The 5 is there to ensure that we don't generate infinite strings.

Size and Error

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generateString = (`generateString'` 5) . simplify
...
generateString' (Star r) n = oneof [generateString' Epsilon n,
            generateString' (Con r (Star r)) (n - 1)]
```

- The 5 is there to ensure that we don't generate infinite strings.

```
generateString' Empty _ = error "cannot generate from empty language"
```

- The `error` is a consequence of not splitting empty and non-empty expressions
- However, the error is only thrown when you do `generateString Empty`.
- That is because of `simplify`

And Simplify?

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- So we did something a bit simpler...

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```
simplify :: RegExp -> RegExp
simplify r | r == simplify' r = r
| otherwise = simplify $ simplify' r
where
simplify' :: RegExp -> RegExp
simplify' Empty = Empty
simplify' Epsilon = Epsilon
simplify' (R xs) = R xs
simplify' (Con Empty Empty) = Empty
simplify' (Con Epsilon Epsilon) = Epsilon
simplify' (Con _ Empty) = Empty
simplify' (Con Empty _) = Empty
simplify' (Con r' Epsilon) = simplify' r'
simplify' (Con Epsilon r') = simplify' r'
simplify' (Con r1 r2) = Con (simplify' r1) (simplify' r2)
simplify' (Union Empty Empty) = Empty
simplify' (Union Empty r') = simplify' r'
simplify' (Union r' Empty) = simplify' r'
simplify' (Union Epsilon Epsilon) = Epsilon
simplify' (Union r1 r2) | r1 /= r2 = Union (simplify' r1) (simplify' r2)
| otherwise = simplify' r1
```

RegExp to DFA

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- But how?
- Recursively!
- First we convert regular expressions to ϵ-NFAs
- Then we just use the power set construction

RegExp to ϵ-NFA: the Base Cases

```
regExpToENFA :: RegExp -> ENFA Int
regExpToENFA Empty    = ENFA [1] [] (\_ _ -> []) [] 1 []
regExpToENFA Epsilon = ENFA [1] [] (\_ _ -> []) [] 1 [1]
regExpToENFA (R [])   = ENFA [1] [] (\_ _ -> []) [] 1 [1]
regExpToENFA (R xs)   = ENFA [0..length xs] (nub xs) delta2 [] 0 [length xs
    ]      where
        delta2 symbol state | state >= length xs || state < 0 = []
        | xs !! state == symbol = [state + 1]
        | otherwise = []
```

RegExp to ϵ-NFA: The Recursion

```
regExpToENFA (Union r1 r2) = regExpToENFA r1 `unionENFA` makeDisjoint (
    regExpToENFA r1) (regExpToENFA r2)
regExpToENFA (Star r) = starENFA (regExpToENFA r)
regExpToENFA (Con r1 r2) = regExpToENFA r1 `concatENFA` makeDisjoint (
    regExpToENFA r1) (regExpToENFA r2)
regExpToENFA (Plus r) = regExpToENFA r `concatENFA` makeDisjoint (
    regExpToENFA r) (starENFA (regExpToENFA r))
```

Combining ϵ-NFA-s: Union

```
unionENFA :: ENFA Int -> ENFA Int -> ENFA Int-- Use only if states are
    disjoint
unionENFA n1 n2 = ENFA states'' alphabet2 delta'' epT start'' accept where
        states'' = start'' : statesENF n1 ++ statesENF n2
        alphabet2 = alphabetENF n1 `union` alphabetENF n2
        delta'' sym st = deltaENF n1 sym st ++ deltaENF n2 sym st
        epT = epTrans n1 ++ epTrans n2 ++ [(start'', startENF n1), (start
            '', startENF n2)]
        start'' = maximum (statesENF n2 ++ statesENF n1) + 1
        accept = acceptstateENF n1 ++ acceptstateENF n2
```

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        alphabet2 = alphabetENF n1 `union` alphabetENF n2
        delta'' sym st = deltaENF n1 sym st ++ deltaENF n2 sym st
        epT = epTrans n1 ++ epTrans n2 ++ [(start'', startENF n1), (start
            '', startENF n2)]
        start'' = maximum (statesENF n2 ++ statesENF n1) + 1
        accept = acceptstateENF n1 ++ acceptstateENF n2
```

- Concat and Star are handled similarly
- Disjoint?

```
ghci> regExpToENFA (Union (R "0") ( R "1"))
ENFA([4,0,1,2,3],"01",["d('0',4) = []","d('0',0) = [1]","d('0',1) =
[]","d('0',2) = []","d('0',3) = []","d('1',4) = []","d('1',0) =
[]","d('1',1) = []","d('1',2) = [3]","d('1',3) =
[]"],[(4,0),(4,2)],4,[1,3])
```

DFA to RegExp

- And now for the fun part!

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```
dfaToRegExp :: Ord a => DFA a -> RegExp
dfaToRegExp dfa = simplify $ regExpUnion [rijk dfaInt (start dfaInt) f (
    length $ states dfaInt) | f <- acceptstate dfaInt ]
        where dfaInt = (makeIntDFA . minimizeDFA) dfa
```

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- The idea is as follows:
- First we make the automaton as small as possible.

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- Then we rename the states a little (to consequetive `Int [1..length $ states dfa]`).

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```

- The idea is as follows:
- First we make the automaton as small as possible.
- Then we rename the states a little (to consequetive `Int [1..length $ states dfa]`).
- Then the magic happens.

Minimising a DFA: Brzozowski's algorithm

- Reversing the DFA
- Cutting out unreachable states
- Reverse it back
- Cut out unreachable states again

```
reverseDFA :: Ord a => DFA a -> DFA [Int]
reverseDFA d = enfaToDFA' $ ENFA sts alph delt ep st acc where
e = singleAccept $ cutDFA d
sts = statesENF e
alph = alphabetENF e
delt sym s = [t | t <- statesENF e, s `elem` deltaENF e sym t]
ep = [(s, t) | (t,s) <- epTrans e ]
st = head $ acceptstateENF e
acc = [startENF e]
```

The Magic

```
rijk :: DFA Int -> Int -> Int -> Int -> RegExp
rijk dfa i j 0 | i == j  = regExpUnion $ Epsilon : labels
| null labels = Empty
| length labels == 1 = head labels
| otherwise = foldr Union (head labels) (tail labels)
where labels =  [R [x] | x <- alphabet dfa, delta dfa x i == j]
rijk dfa i j k = Union (rijk dfa i j (k-1)) (Con (rijk dfa i k (k-1)) (Con
     (Star $ rijk dfa k k (k-1)) (rijk dfa k j (k-1))))
```

- `rijk` stands for R_{ij}^k.
- i and j are indices of states (hence the renaming).
- The k indicates that we only consider states between i and j of index at most k.
- R_{ij}^0 thus covers all paths that do not pass through states.
- $R_{ij}^k := R_{ij}^{k-1} + R_{ik}^{k-1}(R_{kk}^{k-1})^* R_{kj}^{k-1}$

The Magic: continued

$R_{ij}^k := R_{ij}^{k-1} + R_{ik}^{k-1}(R_{kk}^{k-1})^* R_{kj}^{k-1}$:

- Suppose we have a path from i to j that does not visit states higher than k.
- R_{ij}^{k-1} covers the option that the path does not visit k.
- If the path does visit k, then:
- R_{ik}^{k-1} covers the path from i to k.
- $(R_{kk}^{k-1})^*$ covers all (already found) paths from k to itself.
- R_{kj}^{k-1} covers the path from j to k.

The Magic: continued

$R_{ij}^k := R_{ij}^{k-1} + R_{ik}^{k-1}(R_{kk}^{k-1})^* R_{kj}^{k-1}$:

- Suppose we have a path from i to j that does not visit states higher than k.
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- If the path does visit k, then:
- R_{ik}^{k-1} covers the path from i to k.
- $(R_{kk}^{k-1})^*$ covers all (already found) paths from k to itself.
- R_{kj}^{k-1} covers the path from j to k.
- Hence R_{ij}^k is the regular expression covering all paths between i and j not visiting states with index higher than k.

The Magic: the End

recall:

```
dfaToRegExp dfa = simplify $ regExpUnion [rijk dfaInt (start dfaInt) f (
    length $ states dfaInt) | f <- acceptstate dfaInt ]
```

- i is the initial state
- j is an accepting state
- k is the amount of states
- We take the union of R_{ij}^{k} for all accepting states j.

Example

```
zeroStart:: DFA Int
zeroStart = DFA [0,1,2] ['0', '1'] deltazero 0 [1] where
        deltazero:: Symbol -> Int -> Int
        deltazero char st
        | st == 0 && char == '0' = 1
        | st == 0 && char == '1' = 2
        | st == 1 = 1
        | st == 2 = 2
        | otherwise = -1
```

```
ghci> ppRegExp (dfaToRegExp zeroStart)
"("0"|"0"((R e|("0"|"1")))*(R e|("0"|"1")))"
```

Tests

```
"ZeroStart accepts words starting with 0:"
+++ OK, passed 100 tests.
"ZeroStart does not accept other words:"
+++ OK, passed 100 tests.
"Powerset construction yields equivalent DFA for ENFA:"
+++ OK, passed 100 tests.
"DFA accepts words generated by the corresponding RegExp:"
+++ OK, passed 100 tests.
"DFA does not accept words generated by the RegExp corresponding to the
    complement DFA:"
+++ OK, passed 100 tests.
"ENFA corresponding to RegExp accepts words generated from RegExp:"
+++ OK, passed 100 tests.
"Minimize doesn't yield more states"
+++ OK, passed 100 tests.
"cutDFA dfa accepts and rejects the same as dfa"
+++ OK, passed 100 tests.
"The reverse of the reverse is isomorphic to the original"
+++ OK, passed 100 tests.
"Minimize yields an automaton that accepts and rejects the same words:"
+++ OK, passed 100 tests.
"Composition of transitions yields equivalent DFA for DFA:"
+++ OK, passed 100 tests.
```